

Adaptive Advance-Time Control for nFAPI Split under Unpredictable Latency

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Abstract Section (before Introduction) Add a clear placeholder for an abstract:

Purpose: Briefly summarize the problem, method, results, and contribution.

Word count: typically 150—250 words.

Keywords: list 3—6 keywords.

I. Introduction

A. Introduction

In recent years, the Open Radio Access Network (O-RAN) architecture has become popular. O-RAN allows for more flexible network deployments. The Network Functional Application Platform Interface (nFAPI) is an important part of this architecture. nFAPI helps split the O-RAN Distributed Unit (O-DU) into O-DU High and O-DU Low parts. This functional split separates the Media Access Control (MAC) layer and the Physical (PHY) layer. Figure ?? shows the O-RAN nFAPI split architecture.

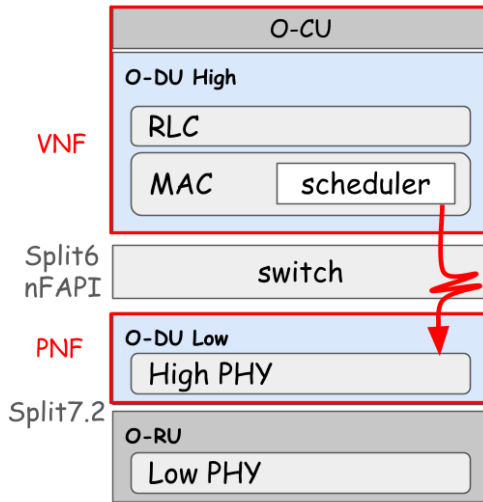


Fig. 1. O-RAN nFAPI functional split architecture.

We define the Virtual Network Function (VNF) to include Layer 2 and Layer 3 operations. The Physical

Network Function (PNF) handles the Layer 1 Physical layer processing. Deploying the VNF and PNF on different machines causes several challenges. Strict slot synchronization is required between these distributed components. Different traffic profiles affect the latency between the VNF and PNF. Figure ?? shows the relationship between different traffic types and the VNF-PNF latency.

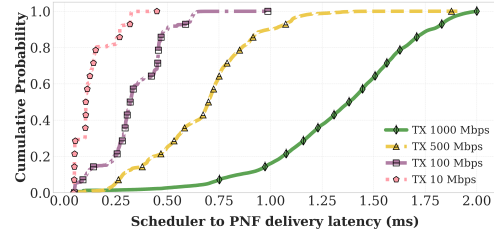


Fig. 2. Observation of latency between VNF scheduler and PNF under different traffic loads.

The system must often face non-ideal network environments. In these environments, the network latency is unpredictable. Figure ?? shows the latency jitter in a non-ideal deployment. Variable latency causes timing issues for the MAC scheduler. Our research shows that dynamic trigger scheduler adjustment is necessary. Using outdated timing information reduces network performance and reliability. This paper proposes an adaptive control method to solve these timing problems.

B. Related Works

C. Contributions

The main contributions of this paper are summarized as follows.

- Contribution 1.
- Contribution 2.
- Contribution 3.

The rest of the paper is organized as follows. Section II defines the system model/architecture considered in this paper. The proposed analytical model/method is given in Section III. Section IV shows the numerical results. Concluding remarks are given in Section VI.

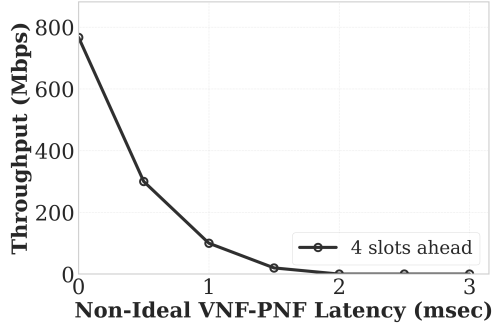


Fig. 3. Impact of unpredictable network latency in a non-ideal environment.

II. System Model/Architecture

TABLE I
nFAPI Delay Management Configuration Parameters

Parameter	Type	Description
timingWindow	uint16_t	Timing window for delay management. See section 3.2.9 of [10]
timingMode	uint8_t	Timing mode for Timing Info. See section 3.2.9 of [10]
timingPeriod	uint8_t	Timing period for Timing Info. See section 3.2.9 of [10]

The following assumptions were made in this paper.

- Assumption 1.
- Assumption 2.
- Assumption 3.

III. Proposed Method

IV. Numerical/Simulation/Experimental Results

Computer simulations/Experiments were conducted to verify xxx (the effectiveness of the proposed analytical model). In the following figures, each point represented the average value of 10^5 samples. In each sample, we collect the statistics of xxx. Symbols and lines were used to show the simulation and analytical results, respectively.

XXX scenarios were investigated. Scenario I was designed to demonstrate/verify the accuracy/effectiveness of Contribution 1.

A. Contribution 1

Fig. 1.1: Purpose X axis: Y axis:

B. Contribution 2

Fig. 2.1: X axis: Y axis:

C. Contribution 3

Fig. 3.1: X axis: Y axis:

References